

Question_9:

Code:

```
1  /*
2  QUESTION_9:
3  Create a Circular Linked List containing numbers from 1 to N.
4  Then split the Circular Linked List into:
5  1. Odd Circular Linked List
6  2. Even Circular Linked List
7  */
8  #include <iostream>
9  using namespace std;
10
11  struct n {
12      int data;
13      n *next;
14  };
15  void insert(int data, n *&head) {
16      n *p = new n{data, NULL};
17
18      if (!head) {
19          head = p;
20          p->next = head;
21      }
22      else {
23          n *t = head;
24
25          while (t->next != head)
26              t = t->next;
27
28          t->next = p;
29          p->next = head;
30      }
31  }
32  void display(n *h) {
33      if (!h)
34          return;
35      n *p = h;
36
37      do {
38          cout << p->data << "=>";
```

```
39         p = p->next;
40     } while (p != h);
41     cout << "[h]";
42 }
43 int main() {
44     int N;
45     cin >> N;
46
47     n *all = NULL;
48     n *odd = NULL;
49     n *even = NULL;
50
51     for (int i = 1; i <= N; i++) {
52         insert(i, all);
53         if (i % 2)
54             insert(i, odd);
55         else
56             insert(i, even);
57     }
58     cout << "Complete linked_list:\n[h]=>";
59     display(all);
60     cout << "\nOdd:\n[h]=>";
61     display(odd);
62     cout << "\nEven:\n[h]=>";
63     display(even);
64     return 0;
65 }
```

Output:

Test Case_1:

```
6
Complete linked_list:
[h]=>1=>2=>3=>4=>5=>6=>[h]
Odd:
[h]=>1=>3=>5=>[h]
Even:
[h]=>2=>4=>6=>[h]
PS C:\Users\Prabin Yadav\Documents
```

Test Case_2:

```
15
Complete linked_list:
[h]=>1=>2=>3=>4=>5=>6=>7=>8=>9=>10=>11=>12=>13=>14=>15=>[h]
Odd:
[h]=>1=>3=>5=>7=>9=>11=>13=>15=>[h]
Even:
[h]=>2=>4=>6=>8=>10=>12=>14=>[h]
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```